# Resolve the Preprocessing Worker Count

## Description

Resolves a safe interpolation worker count from configured workers, a resource profile, an explicit override, and currently available memory.

## Usage

```
resolve_preprocessing_worker_count(
  configured_workers = NULL,
  resource_profile = "shared",
  workers_override = NULL,
  available_memory_bytes = NULL
)
```

## Arguments

| | |
|---|---|
| `configured_workers` | Positive integer worker count from the active analysis profile. |
| `resource_profile` | One of 'shared', 'dedicated', or 'configured'. |
| `workers_override` | Optional explicit positive integer override. A non-empty value has highest priority. |
| `available_memory_bytes` | Optional available-memory estimate in bytes. 'NULL' queries 'ps' when it is available and otherwise skips the memory cap. |

## Details

The shared and dedicated profiles cap concurrency at four and eight workers, respectively. The configured profile preserves the analysis configuration. A conservative memory admission rule reserves eight GiB and allows at most one worker per four remaining GiB. The explicit override bypasses profile caps but not the memory admission rule.

## Value

A single positive integer worker count.